Smart Document Assistant Using RAG with Qwen2.5

Abstract

This project is an AI-powered Smart Document Assistant that enables users to upload PDF documents and ask questions in natural language. The system uses Retrieval-Augmented Generation (RAG) with Qwen2.5 to provide accurate answers grounded in document content.

Problem Statement

Finding information in large PDF documents is time-consuming. Traditional keyword search lacks semantic understanding.

Objectives

Upload PDFs, extract text, split into chunks, generate embeddings, store in FAISS, retrieve relevant chunks, and generate answers using Qwen2.5.

Technology Stack

Frontend: Streamlit. Backend: FastAPI. Document Processing: PyPDF2. Chunking: LangChain RecursiveCharacterTextSplitter. Embeddings: all-MiniLM-L6-v2. Vector DB: FAISS. LLM: Qwen2.5.

Architecture

PDF Upload -> Text Extraction -> Chunking -> Embeddings -> FAISS -> User Question -> Embedding -> Retrieval -> Qwen2.5 -> Answer.

Workflow

1. Upload PDF. 2. Extract text. 3. Create chunks. 4. Generate embeddings. 5. Store in FAISS. 6. Retrieve top-k chunks. 7. Generate answer using Qwen2.5.

Implementation

FastAPI exposes upload and query APIs. Streamlit provides UI. FAISS performs semantic search. Qwen2.5 receives retrieved context and user question.

Advantages

Fast semantic search, accurate grounded responses, scalable, offline-friendly.

Future Scope

Authentication, multi-document search, OCR, multilingual support, cloud deployment.

Conclusion

The project demonstrates an effective SLM-based RAG system using Qwen2.5 for intelligent document question answering.